AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI EXAMINATION

APRIL/ MAY 2018

Physics Paper II

Time: 2 hours 20 minutes Marks: 55

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are FIFTEEN questions. Answer ALL questions. Questions 13, 14 & 15 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue correcting fluid, or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a scientific calculator if you wish.

Q.1. (Total 2 Marks)

The dimension for work done and torque is the same i.e. $[M\,L^{2}\,T^{-2}]$.

Describe the reason for the difference in their S.I. units.

Q.2. (Total 3 Marks)

Show that $2aS = v_f^{\,2} - v_i^{\,2}$ is dimensionally correct.

Q.3. (Total 5 Marks)

An object undergoes a free fall motion. The weight of the object is 10 N and a force of 5 N is applied on the object by the wind blowing from west to east.

a. Sketch a labelled diagram to determine the resultant force. (2 Marks)

b. Calculate the magnitude of the resultant force acting on the object. (3 Marks)

Q.4. (Total 4 Marks)

Identify and discuss the principle used in airbags, truck's arrestor beds and bending your knees when you jump off a chair and land on the ground.

Q.5. (Total 3 Marks)

List any THREE sources of non-conventional energy.

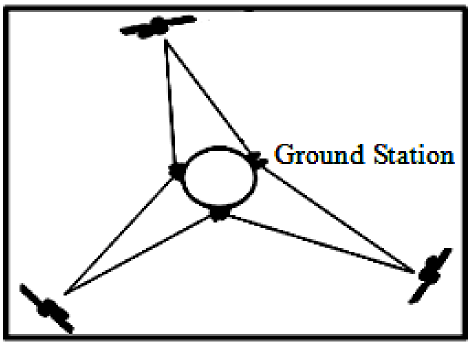
1. ______________________________

2. ______________________________

3. ______________________________

Q.6. (Total 4 Marks)

a. Describe geostationary satellites with respect to the given diagram. (2 Marks)



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b. Artificial gravity can be produced when a manned-satellite revolves around the Earth. Give TWO reasons to support the given statement. (2 Marks)

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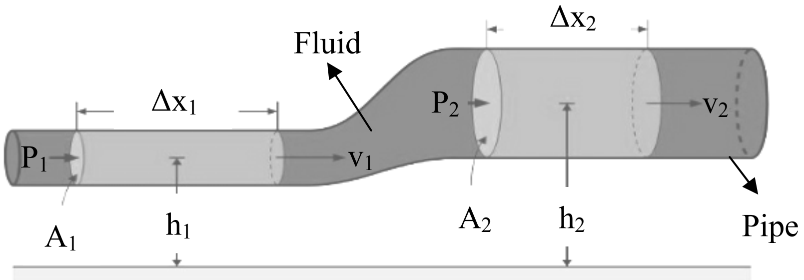
Q.7. (Total 4 Marks)

a. By looking at the given diagram of a pipe, write an appropriate equation for the fluid flow. (1 Mark)



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b. To understand features of fluids in motion, the behaviour of fluids should satisfy THREE conditions. Mention these conditions. (3 Marks)

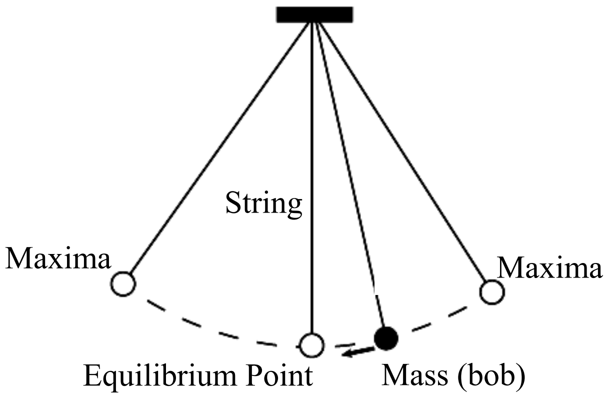
1. ____

2. ____

3. ____

Q.8. (Total 2 Marks)

A simple pendulum consists of a point mass (bob) suspended by a weightless and inextensible string from a fixed support.



Mention TWO factors on which time period of simple pendulum depends.

1. ______________________________

2. ______________________________

Q.9. (Total 3 Marks)

Describe any THREE characteristics of simple harmonic motion.

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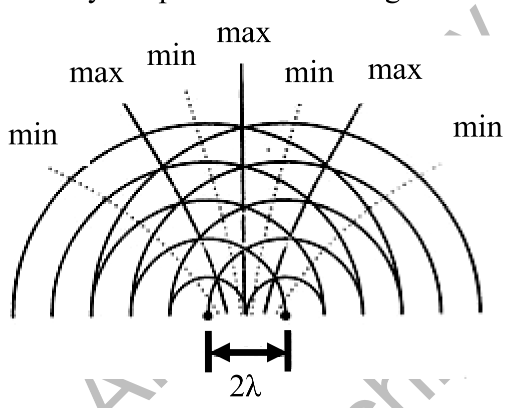
Q.10. (Total 3 Marks)

'A radar works to determine the elevation and speed of an airplane using the principle of Doppler's effect.'

Explain the working of a radar system with the help of the given statement.

Q.11. (Total 4 Marks)

a. Identify the phenomenon of light shown in the given diagram. (1 Mark)



b. Define the phenomenon identified in part (a). (1 Mark)

c. State any TWO conditions required for the occurrence of this phenomenon. (2 Marks)

Q.12. (Total 3 Marks)

A Carnot heat engine with an efficiency of 65% absorbs heat from a reservoir at 590 K. Calculate the exhaust temperature of the engine.

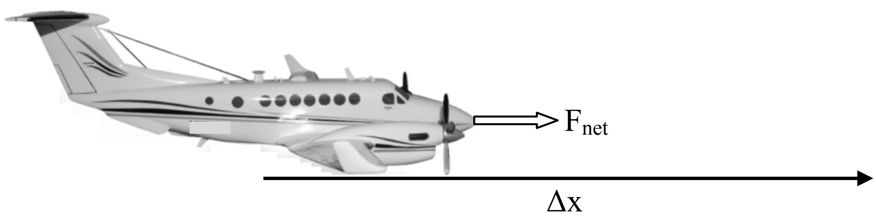
Q.13. (Total 5 Marks)

EITHER

a. Derive mathematical relationship between force, velocity and power applied on a moving object.

OR

b. A constant force (F_{net}) acts on an airplane of mass (m) and it covers a distance (Δx) with an increase in its velocity from (v_i) to (v_f).



Deduce an equation of the work-energy principle for the given situation.

Q.14. (Total 5 Marks)

EITHER

a. Describe Newton's formula for the speed of sound in air and the correction made by Laplace to this formula. (5 Marks)

OR

b.

i. The speed of sound is affected by the variation of pressure, temperature and air density of the atmosphere.

Discuss the given statement by showing mathematical relations and how temperature and density of air affects the speed of sound in gases. (4 Marks)

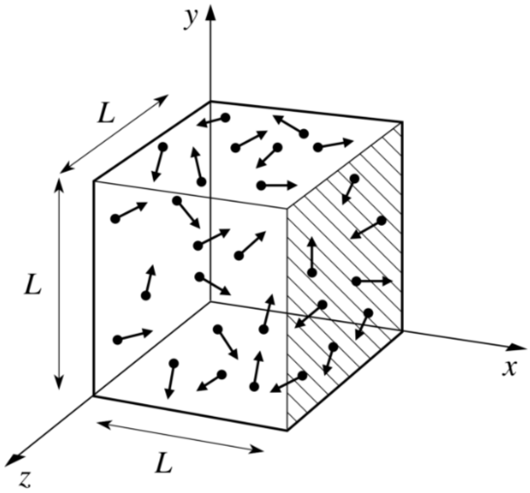
ii. 'The speed of sound is much higher in solids than that of liquids and gases.'

Justify the given statement. (1 Mark)

Q.15. (Total 5 Marks)

EITHER

a. The given figure shows a cube of length (L).



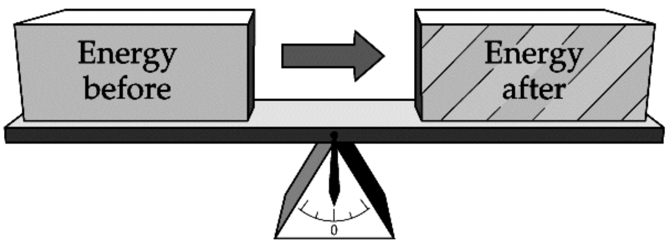
The pressure exerted by the gas molecules in one dimension inside this cube is equal to $P_x = \frac{m}{V} N \overline{v_x^2}$, where ($P_x$) is the pressure in one dimension, (m) is the mass of the gas, (N) is the total number of molecules, ($\overline{v_x^2}$) is the average square speed of the molecules in one dimension.

Deduce an equation for the total pressure exerted by the gas molecules inside the cube. (5 Marks)

OR

b.

i. Interpret the given diagram in light of the first law of thermodynamics. Provide a mathematical equation to support your interpretation. (3 Marks)



ii. Deduce an equation for the first law of thermodynamics in the isothermal process. (2 Marks)

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